

Impact of reference-dependence on a sequential contributions game to public goods

Bachelor thesis

at the

**Faculty of Business, Economic and Social Sciences
of the University of Berne**

supervised by

Prof. Dr. Jean-Michel Benkert

for the award of the title of "Bachelor of Science in Economics"

Author:	David Nhan
Field of study:	Economics
Matriculation no.:	18-928-630
Postal address:	Rosenweg 9 3252 Worben
Email:	david.nhan@students.unibe.ch
Submission date:	11.11.2024

Contents

1	Abstract	3
2	Introduction	4
3	Related Literature	6
4	Model with Reference-Dependency	9
4.1	Model 1: Reference-Dependency for the Second Agent	10
4.2	Model 2: Reference-dependency for the first agent	12
4.3	Model 3: Reference-dependency for both agents	14
4.3.1	Analysis of figure 1	17
4.3.2	Analysis of figure 2,3 and 4	17
5	Model with altruism	17
5.1	Model with warm-glow	19
5.2	Combined Altruism and Warm-Glow Utility Model	20
6	Proof	24
6.1	Model 1: Reference-dependency for the second agent	24
6.2	Model 2: Reference-dependency for the first agent	27
6.3	Model 3: Reference-dependency for both agents	30
6.4	Combined Altruism and Warm-Glow Utility Model	33
7	Conclusion	35
7.1	Changes to the Standard Model	35
7.2	Consequences of the Changes	36
7.3	Summary of Optimal Contribution	37

1 Abstract

This thesis examines the impact of reference-dependent preferences on a sequential contribution game of public goods, expanding on traditional models where free-riding behavior is common. In these standard models, individuals often under-contribute, relying on others to bear the cost. By incorporating reference dependency (where agents wealth acts as a reference point) and warm-glow effects (intrinsic satisfaction from giving), the thesis analyzes how these behavioral factors influence agents contributions.

Different forms of the warm-glow function (linear, concave, convex) are explored, showing how they affect contribution levels in various scenarios. Additionally, a disutility term is introduced to account for the discrepancy between contributions by different agents, further balancing their utilities. The analysis demonstrates that these behavioral components significantly alter the equilibrium, reducing the tendency for free-riding and encouraging more equal contributions between agents. This results in higher overall contributions compared to traditional models, where inefficiencies are more prevalent.

The findings suggest that policy mechanisms exploiting reference points and warm-glow preferences could enhance public goods provision, making the model more applicable to real-world charitable and cooperative behavior.

2 Introduction

My thesis introduces an unexplored aspect of Hal R. Varian's economic model. (1994) which wasn't discussed before in his paper. Varian examines the amount provided to a sequential game by each agent using equilibria and backward induction, demonstrating that the total contributions in a sequential game never exceed the amount in a simultaneous game.

The challenge associated with public goods lies in their tendency to be underprovided in a free market, as individuals may choose to benefit from the good without contributing. This is known as the free-rider problem. Understanding how and why agents decide to contribute to public goods helps in designing better policies and mechanisms to encourage collective action and mitigate issues related to underprovision.

The sequential public goods model is economically interesting because it reflects scenarios where contributions are made sequentially rather than all at once. The order and visibility of contributions of the initial contributions can affect the strategies of the subsequent agent, demonstrating the potential of leadership effects and signaling behavior in economics. Therefore, I believe sequential models provide a clearer perspective of decision-making compared to simultaneous models, where all agents act at the same time without observing others contributions.

In my thesis, I introduce a behavioral component that allows for contributions exceeding the threshold of the public good's value. This economically unusual behavior can be explained through insights from human psychology, particularly how individuals weigh their values and beliefs differently with decisions involving public goods. Unlike traditional economic models that mostly assume rationality and self-interest, real-world behavior often deviates from these assumptions. For instance, people may contribute out of a sense of duty, social pressure, or even altruism, maybe sacrificing personal benefit to support collective welfare. This thesis aims to illustrate these concepts of reference-dependency and its impact on contributions.

I chose to focus on reference-dependency for several reasons: First, it shows the difference tendencies of human behavior, considering how people's preferences influence their decisions. This aspect shows how different individuals prioritize their spending, display consumption patterns, and allocate resources toward public goods. Second, reference-dependency aligns with behavioral economics by being in-between classical economic theory and human behavior. The type of player an agent determines their strategy in contributing to public goods.

To go further into the maybe non-classical decision making, I also introduce a component of altruistic behavior. Altruism provides an explanation for why agents may contribute more than the equilibrium amount, in comparison to the traditional utility-maximizing behavior. In classical economics, agents are expected to act in their self-interest; however, real-world examples show that individuals often act beyond these expectations. These motivations imply that the utility gain from altruism or recognition can outweigh the cost of contributing.

Unlike the work of Varian and studies by Admati and Perry (1991), where contributions in sequential games are capped by rational strategic behavior, my thesis tries to explain scenarios where these constraints can be exceeded. This idea was inspired by my personal experience during a project in which I continued contributing beyond what was necessary because of the satisfaction of the work. This additional effort, where the utility gained from personal satisfaction exceeded the time and resources invested, hopefully offers insight into why agents might go beyond equilibrium contributions.

In summary, by introducing reference-dependency and altruism into the analysis of sequential public goods games, my thesis offers a more comprehensive and realistic approach to understanding why contributions may exceed traditional equilibrium levels. It challenges the conventional wisdom that contributions are strictly limited by strategic rationality and opens up new avenues for explaining cooperative and prosocial behavior in economic models.

3 Related Literature

Sequential Contributions to Public Goods Varian (1994): Varian’s study is foundational for understanding how strategic decision-making unfolds in sequential public goods contributions, revealing the free-rider problem’s persistence and exploring potential subsidy mechanisms. This paper serves as the baseline for my thesis, on which I extend the model by incorporating behavioral elements such as reference dependence, altruistic behavior and warm-glow preferences to modify equilibrium outcomes.

Joint Projects without Commitment Admati and Perry (1991): Admati and Perry’s examination of collaborative projects without commitment highlights strategic trust issues. Where our models overlap are the sunken costs, even without realizing the reward of the public good. The paper furthermore shows an sub game equilibrium, which is not possible in my two rounds contribution model.

Prospect Theory: An Analysis of Decision under Risk Kahneman and Tversky (1979): : Kahneman and Tversky’s prospect theory examines decision-making under uncertainty, contrasting with traditional rational choice theory. Where their expected utility are weighted by non-linear probabilities, my thesis aims to eradicate the uncertainty to introduce a more static idea to the model, where agent’s contributions are visible and influence their choices.

Reference-Dependent Preferences O’Donoghue and Sprenger (2021): O’Donoghue and Sprenger’s work looks into the role of reference points in shaping preferences, supporting my thesis by explaining why agents contributions may vary based on relative starting point, thus affecting the equilibrium in public goods games. Their work focuses more on the expectation-based and gain loss utility reference dependency, whereas mine is more concentrated on the expectation of one agent.

What is Altruism? Khalil (2008): Khalil explores altruism from an economic psychology perspective, emphasizing motivations and social preferences in decision-making. This complements my analysis of how altruistic utility

can encourage agents to exceed typical contribution levels, but without diving deeper into the meaning of altruism and warm-glow.

The Price of Warm Glow Lilley and Slonim (2013): Lilley and Slonim investigate the warm-glow effect, where personal satisfaction from giving changes contributions. We share the idea, that the visibility of the donation plays a significant role as a motivation effect. My thesis extends this idea by considering different forms of warm-glow utility functions, analyzing their impact on agents behavior in sequential public goods games. In addition, no additional costs are bound to the altruistic giving of the agents.

Motivations to Contribute to Public Goods: Beyond Rational Choice Economics Frey and Meier (2004): Frey and Meier argue that intrinsic motivations and social norms change public goods contributions. He argues, that the incentive to contribute must be revised in order to combat free-riding and refusal for cooperation, similar to my focus on behavioral economics elements like warm-glow and reference-dependence that lead to increased contributions differently from normal models.

A Theoretical Analysis of Altruism and Decision Error in Public Goods Games Goeree and Laury (1999): Goeree, Holt, and Laury analyze decision errors and altruism in public goods games. Their findings are about contributions differ, depending on value of the good, number of agents and endowment, whereas my model are assuming a set value and payment of the public good.

Anomalous Behavior in Public Goods Experiments: How Much and Why? Palfrey and Prisbrey (1997): Andreoni reviews deviations from standard economic predictions in public goods experiments, reinforcing the necessity of behavioral extensions like warm-glow and altruism in my model to explain these anomalies.

Appropriating Returns from Innovations in Public Goods Projects Falkinger et al. (2002): Falkinger, Fehr, Gächter, and Winter-Ebmer discuss the allocation of returns from innovations in public goods, highlighting cooperative behavior. My thesis builds on these concepts by showing how

behavioral factors influence contribution levels in non-cooperative settings.

Testing for Altruism and Social Pressure in Charitable Giving

DellaVigna et al. (2012): DellaVigna, List, and Malmendier investigate the influence of altruism and social pressure on giving behavior. This paper's findings provide context for my analysis of warm-glow and altruistic motivations within sequential public goods games.

Optimal Allocation of Public Goods: A Solution to the "Free Rider" Problem

Groves and Ledyard (1977): Groves and Ledyard propose a mechanism to address free-rider challenges in public goods provision by incentivizing optimal contributions. My work differs by focusing on sequential contributions where agents, driven by reference-dependent preferences, may exceed equilibrium contributions due to behavioral factors rather than imposed incentives.

Global Public Goods: A Survey

Buchholz and Sandler (2021): Buchholz and Sandler investigate the importance of public goods, and why policies are made for cooperative provision for the good. This serves as an idea what my finding implicate for decision making of public figures.

Impure Altruism and Donations to Public Goods: A Theory of Warm-Glow Giving

Andreoni (1990): Andreoni's theory gives insight, that pure and impure altruism have different implications for outcomes, therefore changing what policies have to be made. It differs from my perspective, that altruistic behavior, independently from its form is somewhat predictable to the outcome.

Distributive and reciprocal fairness: What can we learn from the heterogeneity of social preferences?

Kamas and Preston (2012): Kamas and Preston put emphasis on inequity averse behavior of agents, showing, high contribution preference for first movers hoping for reciprocal fairness in exchange, which gives basis on my claim of the reference point and social preferences.

4 Model with Reference-Dependency

The general model analyzed is given by:

$$\Pi_i(a_i, a_j) = m_k \left(\sum_s \alpha_s \right) - c(a_i) + \lambda(a_i - a_j)_+ \quad (1)$$

where i represents the agent 1 or agent 2, and k denotes the type of agent i is. In this model, Π_i denotes the utility of agent i . In economics, it is assumed that agents seek to maximize their utility. The function m_k captures an agent's wealth and consumption, aggregated as one term. The cost $c(a_i)$ represents the contributions made by agent i to a public good. The term $\lambda(a_i - a_j)_+$ denotes the reference-dependent utility, which reflects the gains or losses depending on how agent i 's contribution compares to agent j 's.

Assumptions for the Model:

- The full value G for the public good must be contributed for agents to benefit from T , where $T > G$.
- If the total contributions fall short of G , they are not refunded.
- Agents have complete information and can recall contributions made in all previous rounds.
- Each agent knows their own type and the type of the other agent.
- Agents are utility-maximizers.
- The game consists of two rounds, with agent 1 moving first and agent 2 following.
- $\lambda > 0$ to ensure that utility remains non-negative when $w_i = 0$ and $T = G$.

The basic model by H.R. Varian, which serves as the basis for the analysis, is:

$$u_i(G) + x_i = u_i(g_1 + g_2) + w_i - g_i \quad (2)$$

where:

- $G = g_1 + g_2$ is the total value of the public good.
- g_1 and g_2 are contributions from agents 1 and 2, respectively.
- $i \in \{1, 2\}$ denotes the agent index.
- $x_i = w_i - g_i$ represents the agent's consumption for other goods.
- w_i is the initial wealth of the agent.

4.1 Model 1: Reference-Dependency for the Second Agent

In this version, agent 2 incurs a utility loss if their contribution falls short of a reference point, set at $G/2$ for simplicity. The term capturing this effect is:

$$\lambda(g_1 - g_2)_+ = \begin{cases} \lambda(g_1 - g_2), & \text{if } g_1 \geq g_2 \\ 0, & \text{if } g_1 < g_2 \end{cases} \quad (3)$$

This term models the social pressure experienced by agent 2 if their contribution is below a reference point. Unlike agent 1, who follows the basic model without reference-dependence, agent 2's utility depends on how their contribution compares to that of agent 1. The utilities for both agents are:

$$u_1(g_1, g_2) = \begin{cases} w_1 - g_1 + T, & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1, & \text{otherwise} \end{cases} \quad (4)$$

$$u_2(g_1, g_2) = \begin{cases} w_2 - g_2 + T - \lambda(g_1 - g_2)_+, & \text{if } g_1 + g_2 \geq G \\ w_2 - g_2 - \lambda(g_1 - g_2)_+, & \text{otherwise} \end{cases} \quad (5)$$

Agent 1 has the basic model, since Agent 1 does not consider the referenced value. Solving this backwards gives us three cases of contributions for agent 2:

1. **Minimal Contribution (1 unit):** Agent 2 contributes the least, as contributing more decreases their utility.

2. **Maximal Contribution (G - 1 unit):** Agent 2 contributes significantly to avoid disutility from failing to match or exceed the reference point, potentially exceeding the needed total.
3. **Intermediate Contribution:** Agent 2 chooses an amount between the minimum and maximum.

The first two cases can be intuitively shown with numerical examples.

Suppose $G = 10$, $g_1 = 9$, then I have

$$u_2(9, 1) = w_2 - g_2 - \lambda(g_1 - g_2)_+ + T = w_2 - 1 - \lambda 8 + T$$

$$u_2(9, 9) = w_2 - g_2 - \lambda(g_1 - g_2)_+ + T = w_2 - 9 + T$$

Agent 2 contributes more than the needed amount (since agent 1 already invested his maximum) if

$$u_2(9, 9) > u_2(9, 1)$$

$$w_2 - 9 + T > w_2 - 1 - \lambda 8 + T$$

$$\lambda 8 > 8$$

$$\lambda > 1$$

Agent 2 would willingly contribute more than needed for $\lambda > 1$. λ describes the emphasis on contributing more than the other.

What if λ is less than 1, e.g $\lambda = 1/2$?

$$w_2 - 1 - 1/2(9 - 1)_+ + T \leq w_2 - x_2 - 1/2(9 - x_2)_+ + T$$

$$-5 \leq -x_2 - 4.5 + x_2$$

$$0.5x_2 \leq 0.5$$

$$x_2 \leq 1$$

Agent 2 only contributes more than 1 unit (and with that, exceeding the threshold) only if x_2 is bigger than 1, which is never possible with $\lambda = 1/2$,

since agent 2 doesn't care that much. In contrast to Varian's model, the idea of free-riding on the others contribution is eliminated. With $G \geq g_1 + g_1$ and the disutility from contributing less, forces agent 2 to at least match g_1 for a given big λ , as I will show in model 3. I eliminated this idea of free-riding to force participation from both agents, so neither are at a disadvantage.

4.2 Model 2: Reference-dependency for the first agent

Here, only agent 1's utility is affected by how their contribution compares to agent 2's:

$$u_1(g_1, g_2) = \begin{cases} w_1 - g_1 + T - \lambda(g_2 - g_1)_+, & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1, & \text{otherwise} \end{cases} \quad (6)$$

$$u_2(g_1, g_2) = \begin{cases} w_2 - g_2 + T, & \text{if } g_1 + g_2 \geq G \\ w_2 - g_2, & \text{otherwise} \end{cases} \quad (7)$$

Assuming contributions have to be 1 at least to be acknowledged and assuming the threshold can be exceeded.

Agent 2 contributes at maximum $G - g_1$, since he does not have an incentive to give more than needed and at minimum 1.

The following numeric example shows in which case agent 2 would contribute more than the minimum required to reach the goal. In this case I analyze the model numerically with different values of lambda. Assuming that G is equal to 10.

$$\begin{aligned} u_1(9, 1) &= w_1 - 9 + T - \lambda(1 - 9)_+ \\ u_1(1, 9) &= w_1 - 1 + T - \lambda(9 - 1)_+ \\ w_1 - 9 + T - \lambda(1 - 9)_+ &\geq w_1 - 1 + T - \lambda(9 - 1)_+ \\ -9 &\geq -1 - \lambda 8 \\ \lambda 8 &\geq 8 \\ \lambda &\geq 1 \end{aligned}$$

As expected, Agent 1 is willing to provide the larger portion of the contribution if agent 1 really cares about contributing more than the reference point. Now the question remains whether Agent 1 is more willing to contribute the larger portion or something in between the minimum and maximum.

I look at the case in which agent 1 contributes more than the minimum amount in general:

$$\begin{aligned}
u_1(G - g_1, 1) &= w_1 - 1 + T - \lambda((G - 1) - 1)_+ \\
u_1(G - g_1, g_1) &= w_1 - g_1 + T - \lambda((G - g_1) - 1)_+ \\
w_1 - g_1 + T - \lambda((G - g_1) - g_1)_+ &> w_1 - 1 + T - \lambda((G - 1) - 1)_+ \\
-g_1 - \lambda(G - 2g_1)_+ &> -1 - \lambda(G - 2)_+
\end{aligned}$$

in (4.2) there is a case distinction. Since $G \gg 1$ the indicator function on the left side of the inequality is positive only if $g_1 < G/2$. It is equal to zero if $g_1 \geq G/2$.

Case $g_1 < G/2$:

$$\lambda > (g_1 - 1)/(2g_1 + 2)$$

Case $g_1 \geq G/2$:

$$\lambda > (G/2 - 1)/(G + 2)$$

Agent 1 is willing to pay more than the minimum of 1 for $\lambda > (g_1 - 1)/(2g_1 + 2)$, since in the case of $g_1 = G/2$, g_1 can be substituted by $G/2$. This is explained by the following:

If the agent's emphasis of contributing more is bigger, reflected in a large λ , his utility loss from contributing more is smaller than the utility loss from contributing less than the reference point. This is contrary to the free-riding problem, since, in Varian's free-riding problem, agent 1 is willing to contribute first to benefit from free-riding.. In this case, agent 1, who goes first, would willingly contribute more than referenced for a given λ .

4.3 Model 3: Reference-dependency for both agents

I now assume that both agents have a the term. Their utilities are as follows:

$$u_1(g_1, g_2) = \begin{cases} w_1 - g_1 + T - \lambda_1(g_2 - g_1), & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1, & \text{otherwise} \end{cases} \quad (8)$$

$$u_2(g_1, g_2) = \begin{cases} w_2 - g_2 + T - \lambda_2(g_1 - g_2), & \text{if } g_1 + g_2 \geq G \\ w_2 - g_2, & \text{otherwise} \end{cases} \quad (9)$$

I assume that $\lambda_1 = \lambda_2$. Since both agents have the same or at least similar belief, according to H.R Varian, the agent who moves first is at an advantage, since he free-rides on the agent 2's contribution: But now agent 2 could be willing to pay more than needed and therefore maximizing his utility, so that agent 1 could bear a disutility in the end(because he would have contributed less than the reference point post-round). In Varian's version, he has given the idea of bidding to go first. In this case, it could be interesting to analyze, wheter it makes sense to bid to go last. As for the new equilibrium:

Agent 1 would choose his contribution to maximize his utility dependent on his own λ assuming that agent 2 would not contribute more than needed. However, since agent 2 now goes first, agent 2 could contribute more than the threshold, therefore has an advantage. I look now at the utility for agent 2 for some fix g_1 .

$g_1 = 0.5G$: Figure 1 shows, that the utility after the reference point is linearly decreasing, assuming that agent 2 doesn't exceed the threshold. The utility gain or loss before the reference point is dependent on λ :

- for $\lambda > 0.5$ the maximum utility is at the reference point mark.
- for $\lambda = 0.5$ the maximum utility is in $g_1 \in [0.1G, 0.5G]$.

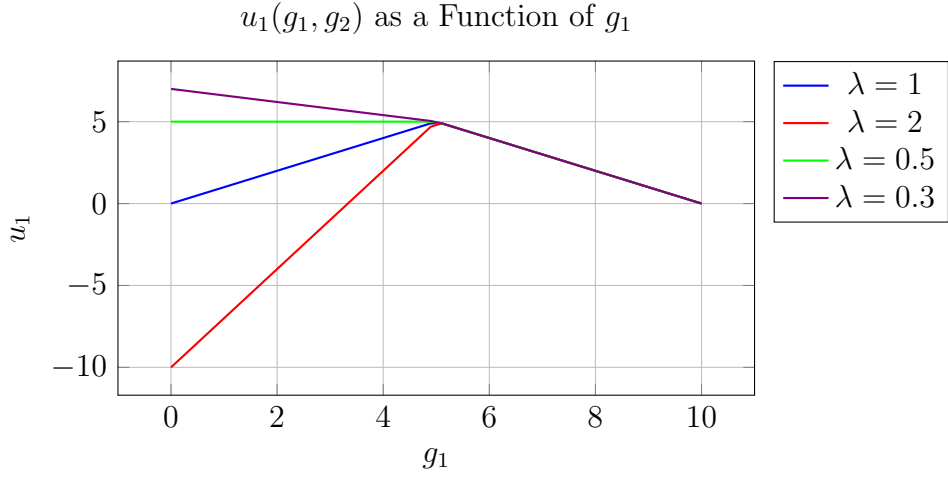


Figure 1: Utility Function $u_1(g_1, g_2)$ as a Function of g_1 for Different λ Values

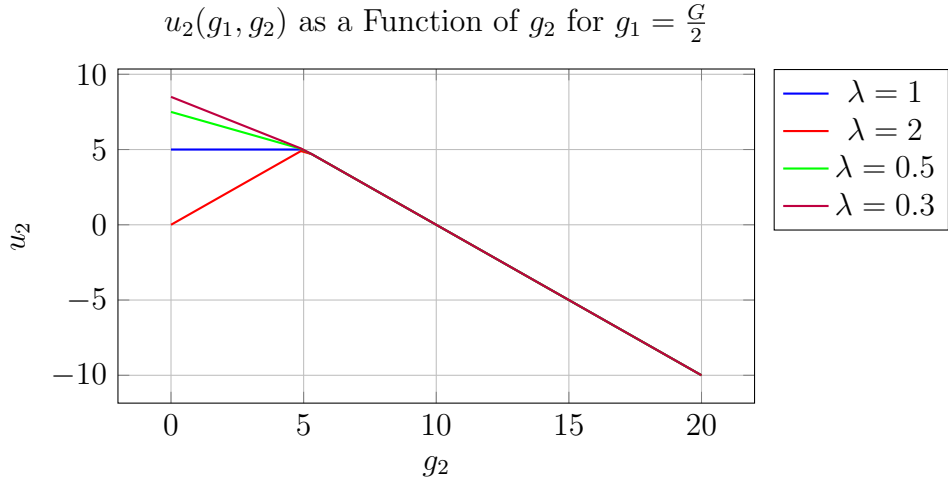


Figure 2: Utility Function $u_2(g_1, g_2)$ as a Function of g_2 for $g_1 = \frac{G}{2}$

- for $\lambda < 0.5$ the utility is decreasing with increasing g_1 until $g_1 = 0.5G$. It further decreases after $g_1 = 0.5G$ but with a different gradient.

The slope of agent 2's utility at contribution $g_2 < 0.5$ is different than agent 1's utility. The slope is flatter for $\lambda \geq 1$ and steeper for $\lambda < 1$. The kink in the utility for agent 2 stays the same as the kink in the utility for agent 2. For every λ it is now more favorable for agent 2 to contribute less than agent 1.

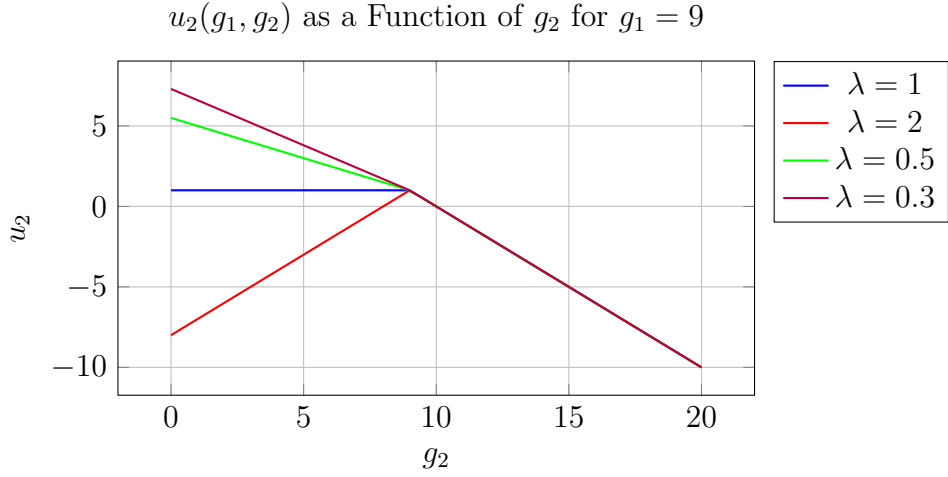


Figure 3: Utility Function $u_2(g_1, g_2)$ as a Function of g_2 for $g_1 = 9$

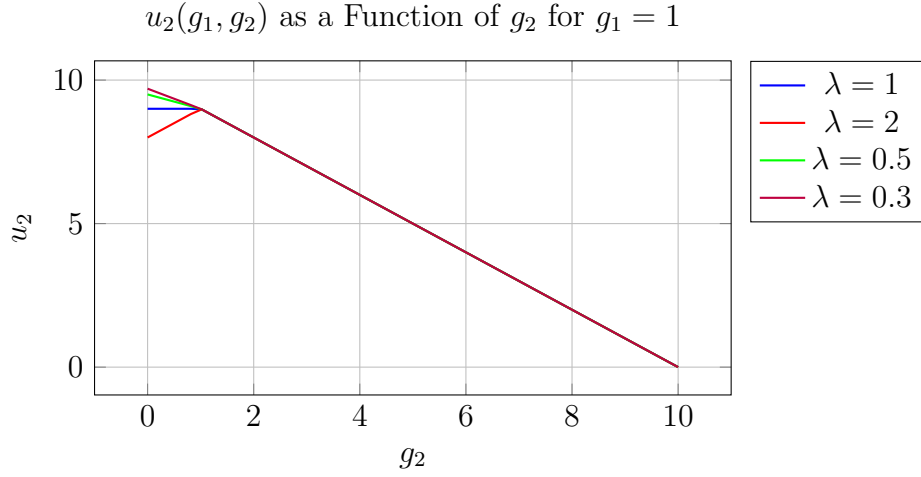


Figure 4: Utility Function $u_2(g_1, g_2)$ as a Function of g_2 for $g_1 = 1$

$g_1 = 0.9G$: The graph shows a similar behavior of the utility for contributions of $g_2 < 0.5G$. The slope is now generally steeper and the maximum utility is lower. The kink is also at a different g_2 , namely at $0.9G$. This is because the λ is fixed and the same for both agents per assumption.

$g_1 = 0.1G$: Similarly to figure 2 it is more favorable to contribute less and the maximum utility is at a higher level.

4.3.1 Analysis of figure 1

I analyze the result by looking at the optima by backward induction. Firstly, the assumption of $\lambda_1 = \lambda_2 = \lambda$ still holds.

For $\lambda \geq 0.5$, g_2 is optimally $0.5G$ which means that $g_1 = 0.5G$. If $\lambda < 0.5$, g_1 is at the minimum, which is $g_1 = 1$ and $g_2 = G - g_1$. This is because agent 1 then has no incentive to contribute more. This leaves u_2 at a lower level. Contrary to Varian, agent 1 is at a disadvantage. For $\lambda > 1$, as shown in Figure 1, agent 2 ideally contributes $0.5G$ as does agent 1.

4.3.2 Analysis of figure 2,3 and 4

Similar to the analysis of figure 1, for some given λ agent 1 and agent 2 are both contributing exactly $0.5G$, namely for $\lambda \geq 1$. For all other $\lambda < 1$, agent 1 and therefore also agent 2 will contribute the minimum amount, namely $g_1 = 0.1G$ and $g_2 = G - g_1 = 0.9G$.

5 Model with altruism

I now introduce a new element to the model. Suppose the utility-gain agent 1 gets is not only dependent on his utility gain, but also on the utility gain from the other agent. Specifically, i introduce a new altruistic parameter $a \geq 0$ so that the new model is as follows:

$$u_{1A} = \begin{cases} u_1(g_1, g_2) + au_2(g_1, g_2) & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1, & \text{otherwise} \end{cases} \quad (10)$$

where u_{1A} is the utility of agent 1, u_{1B} the utility of agent 2 respectively. Now the utility of oneself is dependent on the other agents utility. In contrast to the reference-dependency, the motivation here is more intrinsic. The lambda-term $\lambda(g_2 - g_1)_+$ for agent 1 was either motivated intrinsically by wanting to invest more, since investing more gives agent 1 joy. Or it can

also be explained by social expectations like equal workload/contributions or social pressure not to lack in investments. Now the utility of the agent with the altruistic behavior is not (only) determined by other's expectation, but also determined by the welfare of the other contributor. The reasons for this behavior can be explained by numerous of reasons. The reasons are not limited to, but including:

-Intrinsic motivation: Similar to the previous models, the agent derives satisfaction or fulfilment from helping the other, even at his own costs, as described by Hochman and Rodgers (1969)

-Long-Term Reciprocal Benefits: Assuming there are more than two rounds of contribution. With the added contribution, trust and reciprocal behavior from the other agent can be gained, similar to a tit-for-tat game, described by Axelrod (1984). This can ensure future cooperation for future benefits,

-Social Norms and Reputation: Similar to the intrinsic motivation, but more focus on the self. The agent does this strategically for personal benefits, such as social status.

In the next models, I only focus on the intrinsic motivation. As a start, $\lambda = 1$ is fixed for all subsequent equations. Now I look into the case where agent 1 is altruistic and agent 2 isn't.

For $a = 0$, it stays the same. For $a > 0$ the utility-function becomes more convex. Additionally agent 1's utility has a higher starting point at $g_1 = 1$. This is because, even at max contributions from agent 2, agent 1 would still benefit from the utility gain by agent 2 from the completed project. The optimal g_1 depends on the trade-off between increasing his own utility and increasing agent 2's utility. As a increases and the utility function becomes more convex, agent 1 becomes increasingly inclined to contribute more. For a fixed a namely , $a = 1$ and $\lambda_1 = \lambda_2 = 1$ the utilities g_1, g'_1 are as follows:

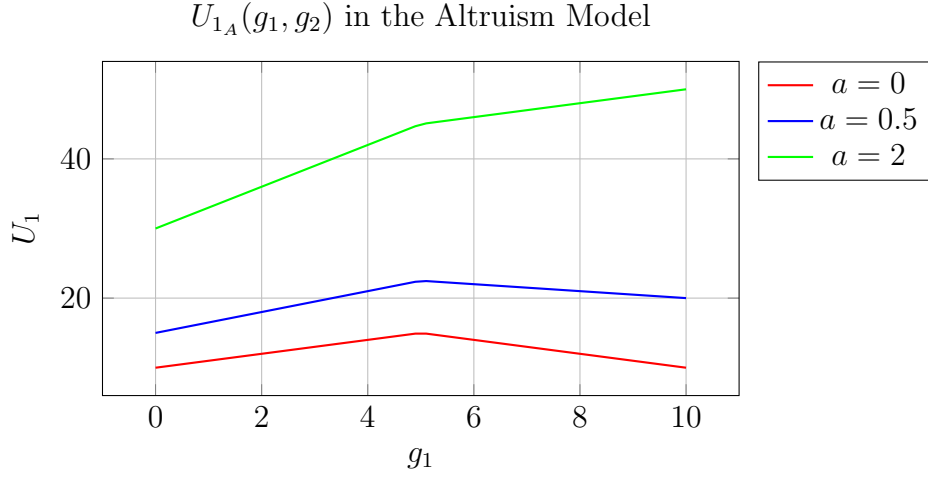


Figure 5: Utility Function $u_{1_A}(g_1, g_2)$ in the Altruism Model for Different Values of a . As shown, for a sufficiently big a the slope after the referenced point is not decreasing anymore, it's increasing. This is due to the fact, that the utility gain of the term $au_2(g_1, g_2)$ is bigger than the cost of contribution of agent 1. This means, that for a sufficiently big a , agent 1 will contribute all of G .

$$u_{1_A} = u_1(g_1, g_2) + u_2(g_1, g_2)u'_{1_A} = u_1(g'_1, g_2) + u_2(g'_1, g_2)$$

the case $u_{1_A} > u'_{1_A}$ is when

$$u_1(g_1, g_2) + u_2(g_1, g_2) > u_1(g'_1, g_2) + u_2(g'_1, g_2) \quad (11)$$

$$w_1 - g_1 + T - \lambda(g_2 - g_1)_+ w_2 - g_2 + T - \lambda(g_1 - g_2)_+ > \quad (12)$$

$$w_1 - g'_1 + T - \lambda(g_2 - g'_1)_+ w_2 - g_2 + T - \lambda(g'_1 - g_2)_+ \quad (13)$$

5.1 Model with warm-glow

In the case of altruism, the utility is dependent on the utilities from both agents. In the warm-glow model only the giving by oneself is relevant. As an example, the contribution I made towards the public good instead of agent 2's contribution gives me satisfaction. Varian has a similar case where he models

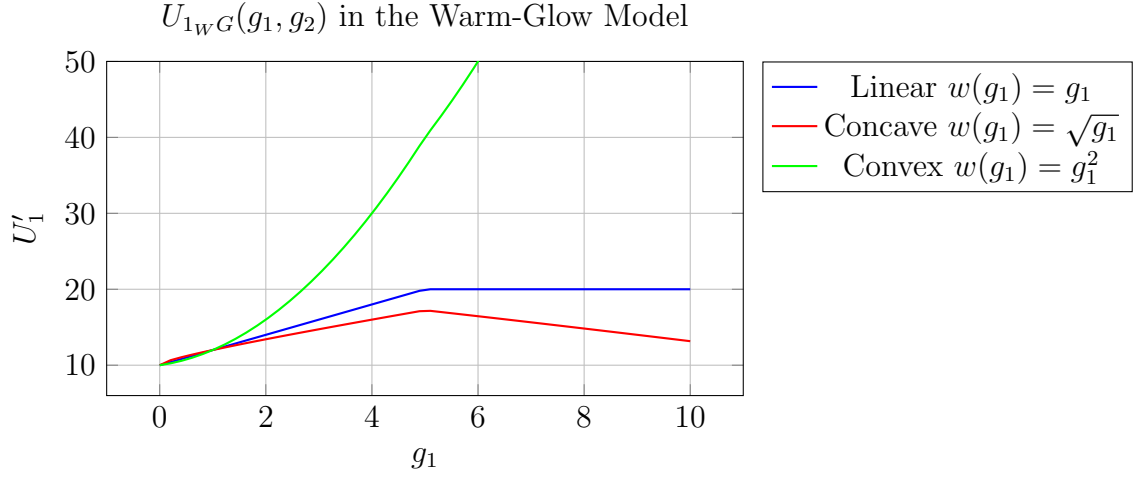


Figure 6: Utility Function $U_{1WG}(g_1, g_2)$ in the Warm-Glow Model with Different Warm-Glow Functions. As shown, the convex function will always result in agent 1 contribution all of G . This is not further analyzed

this warm-glow giving as subsidizing the other player. In this case, the act of giving results in an additional positive utility gain denoted as $\omega(g_1)$. Here, the motivation is purely intrinsic. The different function types I analyze are as follows:

- Linear: $\omega(g_1) = g_1$
- Concave: $\omega(g_1) = \sqrt{g_1}$
- Convex: $\omega(g_1) = (g_1)^2$

So the new warm-glow model looks as follows:

$$u_{1A}(g_1, g_2) = u_1(g_1, g_2) + \omega(g_1)$$

where $\omega(g_1)$ is the warm-glow effect.

5.2 Combined Altruism and Warm-Glow Utility Model

If I combine the altruism as a parameter and the $\omega(g_1)$ into one model it results in this new model:

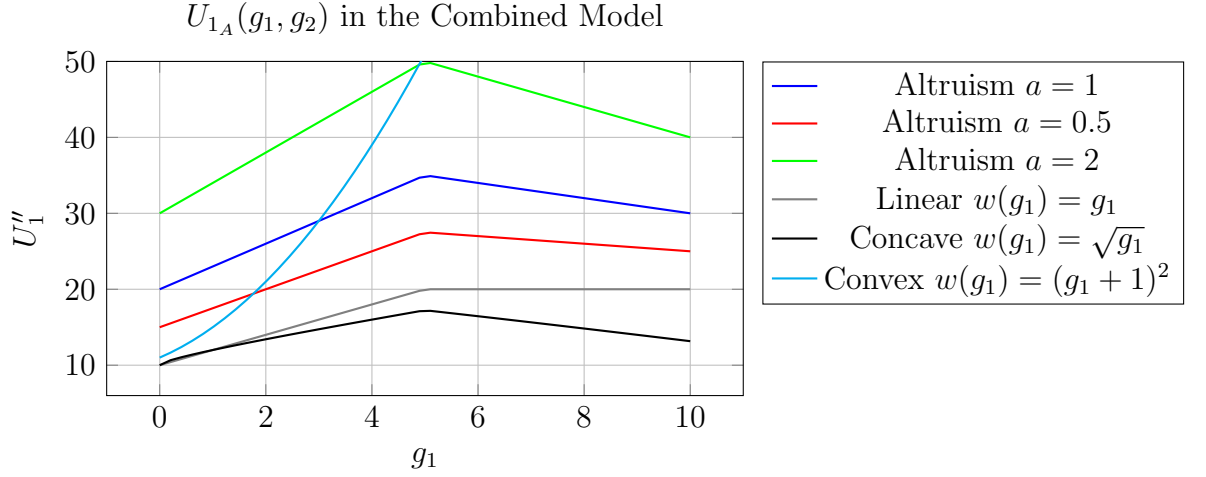


Figure 7: Utility Function $U_{1A}(g_1, g_2)$ in the Combined Altruism and Warm-Glow Model. For the first 3 cases a linear function is used. For the last 3 cases $a = 0$

$$u_{1A+WG}(g_1, g_2) = \begin{cases} u_1(g_1, g_2) + au_2(g_1, g_2) + \omega(g_1), & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1 + \omega(g_1), & \text{otherwise} \end{cases}$$

This utility function has three key factors for agent 1:

1. *Direct utility* from the public good outcome, $u_1(g_1, g_2)$,
2. *Altruistic utility* from agent 2's welfare, $au_2(g_1, g_2)$,
3. *Warm-glow utility* $\omega(g_1)$, reflecting personal satisfaction from giving.

Analysis The base case where $a = 0$ and $\omega(g_1) = g_1$ reduces the model to the basic model without altruism and warm-glow effect. Furthermore, the case with convex warm-glow function is not discussed as it is clear as seen in the graph.

The different altruism variable a defines the overall utility for the agent, as the agent will already benefit even with zero contributions. a defines at which level the utility starts. Intuitively this is explained as follows: Assuming that both agent are willing to contribute to fulfill G , agent 1 will already benefit

at $g_1 = 0$ since then $g_2 = G$ and $a > 0$ which leads to $au_2(g_1, g_2) > 0$, since agent 2 will benefit from T . After the reference point, the slope is flatter than before the kink. This is explained by the fact that after this point, agent 2 profits more, since agent 2 would need to pay less. This lessens the utility loss for agent 1 since, $a > 0$.

The function $\omega(g_1)$ defines the course of the utility function in the graph. With increasing convexity agent 1 has a higher utility gain from contributing more.

Optimal Contribution To find agent 1's optimal contribution g_1^* , I balance the three components of utility: own wealth, altruism, and warm-glow. The optimal g_1^* is found by solving the first-order condition with the following assumptions: $-a = 1$ since a only defines overall utility

$$-\omega(g_1) = g_1 \text{ linear } -g_1 + g_2 > G$$

If I combine altruism as a parameter and the warm-glow function $\omega(g_1)$ into one model, I get the following utility function for agent 1:

$$u_{1A+WG}(g_1, g_2) = \begin{cases} u_1(g_1, g_2) + au_2(g_1, g_2) + \omega(g_1), & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1 + \omega(g_1), & \text{otherwise} \end{cases}$$

This utility function incorporates three key components for agent 1:

1. *Direct utility* from the public good outcome, $u_1(g_1, g_2)$,
2. *Altruistic utility* from agent 2's welfare, $au_2(g_1, g_2)$,
3. *Warm-glow utility* $\omega(g_1)$, which reflects personal satisfaction from giving.

Analysis Let's consider the case where $a = 1$ and $\omega(g_1) = g_1$ (linear warm-glow). The direct utility of agent 1 is $u_1(g_1, g_2) = w_1 - g_1 + T$, where T represents the total benefit from the public good when the project is completed. Similarly, the altruistic utility is $u_2(g_1, g_2) = w_2 - g_2 + T$, where agent 1 benefits from agent 2's welfare.

To simplify, assume $g_1 + g_2 \geq G$, meaning that the public good is funded. The warm-glow utility is $\omega(g_1) = g_1$.

The combined utility function is:

$$u_{1A+WG}(g_1, g_2) = (w_1 - g_1 + T) + (w_2 - g_2 + T) + g_1$$

where $g_2 = G - g_1$ because the total contribution satisfies $g_1 + g_2 = G$.

Substituting $g_2 = G - g_1$ into the utility function:

$$u_{1A+WG}(g_1) = (w_1 - g_1 + T) + (w_2 - (G - g_1) + T) + g_1$$

Simplifying:

$$u_{1A+WG}(g_1) = w_1 - g_1 + T + w_2 - G + g_1 + T + g_1$$

$$u_{1A+WG}(g_1) = w_1 + w_2 - G + 2T + g_1$$

Optimal Contribution To find agent 1's optimal contribution g_1^* , I need to maximize $u_{1A+WG}(g_1)$. I do this by taking the derivative of the utility function with respect to g_1 and setting it equal to zero.

$$\frac{du_{1A+WG}(g_1)}{dg_1} = 0$$

Since $u_{1A+WG}(g_1) = w_1 + w_2 - G + 2T + g_1$, the derivative is:

$$\frac{du_{1A+WG}(g_1)}{dg_1} = 1$$

This means that the utility is strictly increasing with respect to g_1 , so the optimal contribution for agent 1 is to contribute as much as possible, i.e., $g_1^* = G$.

The optimal contribution for agent 1 is $g_1^* = G$, meaning that agent 1 will fully contribute to the public good when $a = 1$ and $\omega(g_1) = g_1$.

6 Proof

6.1 Model 1: Reference-dependency for the second agent

In this model, agent 2 experiences a utility loss if their contribution, g_2 , is less than the reference point set by agent 1's contribution, g_1 . The disutility term is modeled as:

$$\lambda(g_1 - g_2)_+ = \begin{cases} \lambda(g_1 - g_2), & \text{if } g_1 \geq g_2, \\ 0, & \text{if } g_1 < g_2. \end{cases}$$

This disutility reflects a situation where agent 2 cares about not contributing less than agent 1 due to social pressure or expectations.

Utility Functions for Both Agents

Agent 1's utility is:

$$u_1(g_1, g_2) = \begin{cases} w_1 - g_1 + T, & \text{if } g_1 + g_2 \geq G, \\ w_1 - g_1, & \text{if } g_1 + g_2 < G. \end{cases}$$

Agent 2's utility incorporates the disutility term and is:

$$u_2(g_1, g_2) = \begin{cases} w_2 - g_2 + T - \lambda(g_1 - g_2)_+, & \text{if } g_1 + g_2 \geq G, \\ w_2 - g_2 - \lambda(g_1 - g_2)_+, & \text{if } g_1 + g_2 < G. \end{cases}$$

Agent 2's Contribution Choices

I now analyze agent 2's optimal contribution, g_2 , by considering the following three cases: contributing the least, contributing the most, or contributing somewhere in between.

Case 1: Agent 2 Contributes the Minimum $g_2 = \varepsilon$ Let's first consider the situation where agent 2 contributes the smallest possible amount, denoted as $g_2 = \varepsilon$, where $\varepsilon > 0$. In this case, their utility becomes:

$$u_2(g_1, \varepsilon) = \begin{cases} w_2 - \varepsilon + T - \lambda(g_1 - \varepsilon), & \text{if } g_1 + \varepsilon \geq G, \\ w_2 - \varepsilon - \lambda(g_1 - \varepsilon), & \text{if } g_1 + \varepsilon < G. \end{cases}$$

If agent 1 contributes a large amount relative to G , i.e., g_1 is close to or exceeds G , agent 2 might be able to free-ride with a small contribution, but the disutility from contributing less than agent 1, $\lambda(g_1 - \varepsilon)$, reduces agent 2's utility. Thus, the utility loss due to social pressure might outweigh the benefits of contributing little.

Case 2: Agent 2 Contributes the Maximum $g_2 = G - g_1$ In the second case, agent 2 might contribute enough to just meet the threshold G , i.e., $g_2 = G - g_1$. Here, agent 2's utility becomes:

$$u_2(g_1, G - g_1) = w_2 - (G - g_1) + T.$$

Notice that in this case, agent 2 avoids any disutility from contributing less than agent 1 because they are matching or exceeding the total required contribution for the public good. Therefore, the disutility term $\lambda(g_1 - g_2)_+$ does not apply when $g_2 \geq g_1$, making this a feasible option if the disutility from contributing less than agent 1 is significant enough.

Case 3: Agent 2 Contributes an Intermediate Amount $g_2 \in (\varepsilon, G - g_1)$ In the third case, agent 2 might contribute an intermediate amount such that $g_2 \in (\varepsilon, G - g_1)$. Their utility is:

$$u_2(g_1, g_2) = \begin{cases} w_2 - g_2 + T - \lambda(g_1 - g_2), & \text{if } g_1 + g_2 \geq G, \\ w_2 - g_2 - \lambda(g_1 - g_2), & \text{if } g_1 + g_2 < G. \end{cases}$$

In this case, agent 2 still suffers the disutility from contributing less than agent 1, but the utility loss will depend on the strength of λ and the size of the gap between g_1 and g_2 .

Optimal Contribution for Agent 2

I now compare the utility values across the three cases to determine the optimal contribution for agent 2.

Comparison of Case 1 and Case 2: Agent 2 will prefer to contribute more than the minimum if the utility from contributing the maximum (Case 2) exceeds the utility from contributing the minimum (Case 1). Formally:

$$u_2(g_1, G - g_1) > u_2(g_1, \varepsilon).$$

Substituting the utility expressions:

$$w_2 - (G - g_1) + T > w_2 - \varepsilon + T - \lambda(g_1 - \varepsilon),$$

which simplifies to:

$$-(G - g_1) > -\varepsilon - \lambda(g_1 - \varepsilon).$$

For sufficiently large λ , the disutility term $\lambda(g_1 - \varepsilon)$ becomes significant, making agent 2 prefer contributing close to or matching g_1 . This shows that if the disutility from contributing less than agent 1 is strong (i.e., large λ), agent 2 will contribute more.

Comparison of Case 1 and Case 3: Similarly, agent 2 will prefer contributing an intermediate amount g_2 over the minimum ε if:

$$u_2(g_1, g_2) > u_2(g_1, \varepsilon),$$

or:

$$w_2 - g_2 + T - \lambda(g_1 - g_2) > w_2 - \varepsilon + T - \lambda(g_1 - \varepsilon).$$

This can be rearranged as:

$$-(g_2 + \lambda(g_1 - g_2)) > -(\varepsilon + \lambda(g_1 - \varepsilon)).$$

For sufficiently large g_2 , agent 2 might avoid the disutility term while still making a moderate contribution. However, if λ is strong enough, agent 2 is incentivized to match or exceed agent 1's contribution to eliminate the disutility altogether, making Case 2 the optimal choice.

6.2 Model 2: Reference-dependency for the first agent

In this model, agent 1 experiences a utility loss if their contribution, g_1 , is less than the reference point set by agent 2's contribution, g_2 . The disutility term is modeled as:

$$\lambda(g_2 - g_1)_+ = \begin{cases} \lambda(g_2 - g_1), & \text{if } g_2 \geq g_1, \\ 0, & \text{if } g_2 \leq g_1. \end{cases}$$

This disutility reflects agent 1's concern about contributing less than agent 2 due to social pressure or expectations.

Utility Functions for Both Agents

Agent 1's utility is:

$$u_1(g_1, g_2) = \begin{cases} w_1 - g_1 + T - \lambda(g_2 - g_1)_+, & \text{if } g_1 + g_2 \geq G, \\ w_1 - g_1, & \text{if } g_1 + g_2 < G. \end{cases}$$

Agent 2's utility is:

$$u_2(g_1, g_2) = \begin{cases} w_2 - g_2 + T, & \text{if } g_1 + g_2 \geq G, \\ w_2 - g_2, & \text{if } g_1 + g_2 < G. \end{cases}$$

Agent 1's Contribution Choices

I now analyze agent 1's optimal contribution, g_1 , considering their disutility from contributing less than agent 2. There are two key cases to explore: when agent 1 contributes less than g_2 (and thus incurs disutility) and when they contribute more than or equal to g_2 (no disutility).

Case 1: Agent 1 Contributes Less Than g_2 In this case, agent 1 contributes $g_1 < g_2$, so the disutility term $\lambda(g_2 - g_1)$ applies. Agent 1's utility becomes:

$$u_1(g_1, g_2) = w_1 - g_1 + T - \lambda(g_2 - g_1).$$

To avoid the disutility, agent 1 has an incentive to increase their contribution to match or exceed g_2 , especially when λ is large, reflecting strong concern about contributing less than agent 2.

Case 2: Agent 1 Contributes More Than or Equal to g_2 In this case, agent 1 contributes $g_1 \geq g_2$, so the disutility term $\lambda(g_2 - g_1)_+$ vanishes. Agent 1's utility becomes:

$$u_1(g_1, g_2) = w_1 - g_1 + T.$$

There is no penalty from the disutility term, and agent 1 only incurs the cost of their own contribution. Agent 1's utility is higher compared to Case 1 if they contribute more than g_2 , depending on the size of λ .

Optimal Contribution for Agent 1

I now compare the two cases to determine the conditions under which agent 1 would contribute more or less than g_2 .

Comparison of Contributions: Agent 1 will prefer contributing more than or equal to g_2 if:

$$u_1(g_1 \geq g_2, g_2) > u_1(g_1 < g_2, g_2).$$

Substituting the utility expressions:

$$w_1 - g_1 + T > w_1 - g_1 + T - \lambda(g_2 - g_1),$$

which simplifies to:

$$0 > -\lambda(g_2 - g_1),$$

or:

$$g_1 \geq g_2.$$

Thus, agent 1 avoids the disutility by matching or exceeding agent 2's contribution.

General Case Analysis

I can analyze the general case to determine when agent 1 is willing to contribute more than the minimum. Suppose agent 1 considers two contribution levels: the minimum contribution $g_1 = \varepsilon$ (just enough to participate) and a larger amount $g_1 \geq g_2$.

For $g_1 = \varepsilon$, agent 1's utility becomes:

$$u_1(\varepsilon, g_2) = w_1 - \varepsilon + T - \lambda(g_2 - \varepsilon).$$

For $g_1 \geq g_2$, agent 1's utility becomes:

$$u_1(g_1, g_2) = w_1 - g_1 + T.$$

Comparing the two:

$$w_1 - g_1 + T > w_1 - \varepsilon + T - \lambda(g_2 - \varepsilon),$$

which simplifies to:

$$g_1 < \varepsilon + \lambda(g_2 - \varepsilon).$$

Thus, agent 1 will contribute more than the minimum if the disutility from contributing less than g_2 (governed by λ) is large enough.

Case Distinction Based on λ

To formalize agent 1's decision, I look at two possible scenarios depending on the value of λ .

Case $g_1 < G/2$: When $g_1 < G/2$, agent 1 is likely to contribute more to avoid the disutility, since contributing less than half of the total public good threshold G would make the disutility term significant. Specifically, agent 1 prefers contributing more if:

$$\lambda(2g_1 - 2) > -1 + g_1,$$

which simplifies to:

$$\lambda > \frac{g_1 - 1}{2g_1 + 2}.$$

Case $g_1 \geq G/2$: When $g_1 \geq G/2$, the disutility term vanishes, and agent 1 has no strong incentive to contribute significantly more than necessary. In this case, agent 1's decision is based purely on avoiding the cost of their contribution, with the disutility term not applying.

Agent 1 contributes more than the minimum when λ is large enough to outweigh the cost of contributing less than the reference point set by agent 2.

6.3 Model 3: Reference-dependency for both agents

In this model, I assume that both agents have reference-dependency in their utilities. Their utility functions are defined as follows:

$$u_1(g_1, g_2) = \begin{cases} w_1 - g_1 + T - \lambda_1(g_2 - g_1), & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1, & \text{otherwise} \end{cases}$$
$$u_2(g_1, g_2) = \begin{cases} w_2 - g_2 + T - \lambda_2(g_1 - g_2), & \text{if } g_1 + g_2 \geq G \\ w_2 - g_2, & \text{otherwise} \end{cases}$$

Here, g_1 and g_2 represent the contributions of agent 1 and agent 2, respectively, towards a public good. The parameter G is the threshold that must be met for the public good to be provided. The parameter T represents the value of the public good, and λ_1, λ_2 capture the degree of reference-dependency for each agent.

In this model, I assume that both agents have the same reference-dependency, i.e., $\lambda_1 = \lambda_2 = \lambda$. The agents utilities depend not only on their contributions but also on how their contributions compare to each other's.

Given the reference-dependency of both agents, the agent who moves second might have an advantage since they can adjust their contribution after observing the other agent's contribution. I will now analyze the strategic interactions between the agents in three cases, considering different contributions of agent 1.

Case 1: $g_1 = \frac{G}{2}$ In this case, agent 1 contributes half of the required total, $g_1 = \frac{G}{2}$. The utility function for agent 2 is as follows:

$$u_2 \left(g_1 = \frac{G}{2}, g_2 \right) = \begin{cases} w_2 - g_2 + T - \lambda \left(\frac{G}{2} - g_2 \right), & \text{if } g_1 + g_2 \geq G \\ w_2 - g_2, & \text{otherwise} \end{cases}$$

Agent 2's utility depends on how their contribution compares to agent 1's. For $g_2 \geq \frac{G}{2}$, the public good is provided and agent 2 suffers a disutility proportional to $g_1 - g_2$ (the difference between their contributions). If $\lambda > 1$, agent 2 prefers to contribute less than agent 1 to avoid the cost of contributing more. For $\lambda < 1$, the disutility is less severe, and agent 2 may be willing to contribute more.

The slope of agent 2's utility, $u_2(g_1 = \frac{G}{2}, g_2)$, is determined by the parameter λ . For $\lambda \geq 1$, the slope is flatter, implying that contributing more than agent 1 leads to a significant reduction in utility. For $\lambda < 1$, the slope is steeper, suggesting that contributing more has a less severe effect on agent 2's utility.

In this case, agent 2 may strategically contribute less than agent 1 if $\lambda \geq 1$, as the disutility from contributing more is high. Conversely, if $\lambda < 1$, agent 2 might be more willing to match or exceed agent 1's contribution.

Case 2: $g_1 = xG$ **with** $0.5 < x < 1$ In this scenario, agent 1 contributes more than half of the required total, i.e., $g_1 = xG$ where $0.5 < x < 1$. The utility function for agent 2 becomes:

$$u_2(g_1 = xG, g_2) = \begin{cases} w_2 - g_2 + T - \lambda(xG - g_2), & \text{if } g_1 + g_2 \geq G \\ w_2 - g_2, & \text{otherwise} \end{cases}$$

Here, agent 2's utility depends on the difference between agent 1's contribution and their own. Since agent 1 has contributed a large share of the total, agent 2 might find it advantageous to contribute less than agent 1, especially when $\lambda \geq 1$, as the disutility from exceeding agent 1's contribution increases.

In this case, the utility function for agent 2 shows that the slope is steeper when g_1 is large. The agent's utility decreases more rapidly as they increase their contribution beyond what is necessary to meet the threshold G . Thus, agent 2 is incentivized to minimize their contribution and rely on agent 1 to bear a greater share of the cost.

For values of $\lambda \geq 1$, agent 2 will prefer to free-ride on agent 1's larger contribution. If $\lambda < 1$, agent 2 may still contribute a substantial amount, but the strategic interaction will depend on how much weight agent 2 places on the reference-dependency.

Case 3: $g_1 = yG$ **with** $0 < y < 0.5$ In this final case, agent 1 contributes less than half of the required total, i.e., $g_1 = yG$ where $0 < y < 0.5$. Agent 2's utility function becomes:

$$u_2(g_1 = yG, g_2) = \begin{cases} w_2 - g_2 + T - \lambda(yG - g_2), & \text{if } g_1 + g_2 \geq G \\ w_2 - g_2, & \text{otherwise} \end{cases}$$

In this case, agent 2 faces a similar situation as agent 1 in the earlier cases. Since agent 1 has contributed less than half of the required total, agent 2 is likely to bear a larger share of the burden. However, the steepness of the utility function still depends on λ .

For small contributions by agent 1, the utility function for agent 2 may exhibit a more gradual slope if $\lambda \geq 1$, indicating that agent 2 can achieve higher utility by contributing more and meeting the threshold. However, for $\lambda < 1$, the slope is steeper, suggesting that agent 2's utility decreases rapidly as they contribute more than necessary.

Agent 2 may be more willing to contribute a larger share when g_1 is small, especially if $\lambda \geq 1$. However, for $\lambda < 1$, agent 2 may still prefer to contribute less,

In this model, the strategic interaction between the agents is heavily influenced by their reference-dependent utilities. The agent who moves second may have an advantage in some cases, as they can adjust their contribution to maximize their utility based on the contribution of the first agent. The exact equilibrium depends on the value of λ , which determines how much disutility each agent experiences from contributing more or less than the other.

6.4 Combined Altruism and Warm-Glow Utility Model

In the combined altruism and warm-glow utility model, the utility function for agent 1 is given by:

$$u_{1A+WG}(g_1, g_2) = \begin{cases} u_1(g_1, g_2) + au_2(g_1, g_2) + \omega(g_1), & \text{if } g_1 + g_2 \geq G \\ w_1 - g_1 + \omega(g_1), & \text{otherwise} \end{cases}$$

where:

- $u_1(g_1, g_2)$ is agent 1's direct utility from the public good.
- $u_2(g_1, g_2)$ is agent 2's utility, which is weighted by a in agent 1's utility.
- $\omega(g_1)$ is agent 1's warm-glow utility from contributing.

- G is the total contribution required for the public good.
- $g_1 + g_2 = G$ ensures that the public good is funded.

Let us assume $\omega(g_1) = g_1$ (linear warm-glow) and set $a = 1$ for simplicity.

For the case where $g_1 + g_2 \geq G$, I substitute $g_2 = G - g_1$ into the utility function:

$$u_{1A+WG}(g_1) = (w_1 - g_1 + T) + a(w_2 - (G - g_1) + T) + \omega(g_1)$$

Substituting $a = 1$ and $\omega(g_1) = g_1$:

$$u_{1A+WG}(g_1) = (w_1 - g_1 + T) + (w_2 - G + g_1 + T) + g_1$$

Simplifying:

$$u_{1A+WG}(g_1) = w_1 + w_2 - G + 2T + g_1$$

To find the optimal contribution g_1^* , I take the derivative of the utility function with respect to g_1 :

$$\begin{aligned} \frac{du_{1A+WG}(g_1)}{dg_1} &= \frac{d}{dg_1}(w_1 + w_2 - G + 2T + g_1) \\ \frac{du_{1A+WG}(g_1)}{dg_1} &= 1 \end{aligned}$$

Since the derivative is a constant 1, the utility is strictly increasing with respect to g_1 . Therefore, agent 1 maximizes their utility by contributing the maximum amount, i.e., $g_1^* = G$.

The optimal contribution for agent 1 is $g_1^* = G$, meaning that agent 1 will fully contribute to the public good when $a = 1$ and $\omega(g_1) = g_1$ (linear warm-glow).

7 Conclusion

In this thesis, I explore how reference dependency, altruism, and warm-glow modify the contributions of agents in a public goods setting, particularly compared to the standard model of sequential contributions from Varian. The analysis incorporates various psychological and behavioral factors into a traditional economic framework, leading to these insights into agents behaviors:

7.1 Changes to the Standard Model

Incorporation of Altruism : *Altruistic Utility*: The altruism of agent 1 is modeled by introducing a parameter a into their utility function. This parameter reflects how much agent 1 cares about agent 2's welfare. In the original sequential contribution model, agents only care about their own contributions and the resulting public good. However, by adding altruism, agent 1 now derives utility from agent 2's welfare, represented as $u_2(g_1, g_2)$. *Impact on Contributions*: In the altruistic model, when $a > 0$, agent 1 is willing to contribute more because they not only value their own utility but also the utility of agent 2. As a result, the contribution burden may shift more towards agent 1, potentially mitigating free-riding tendencies by agent 2.

Warm-Glow Utility : *Warm-Glow Effect*: The warm-glow model is introduced to capture the intrinsic satisfaction agents receive from making contributions to the public good. This is modeled by a function $\omega(g_1)$, with different functional forms (linear, concave, convex). This warm-glow component is independent of the final public good outcome. *Impact on Contributions*: With a strong warm-glow effect, agents are incentivized to contribute more, as they derive satisfaction from the act of giving itself. This can further reduce the tendency for agent 1 to free-ride or delay their contribution.

Reference Dependency and Disutility : *Disutility from Imbalance*: Reference dependency is introduced by adding a disutility term to account for

perceived imbalances in contributions between agents. Agent 1 dislikes contributing less than agent 2 and incurs a penalty when $g_2 > g_1$. This disutility is modeled using $\lambda(g_2 - g_1)_+$. *Impact on Contributions:* The introduction of this disutility term encourages agent 1 to match or exceed agent 2's contribution to avoid the disutility. This shifts the equilibrium of the game, as agent 1 becomes more responsive to the actions of agent 2, promoting higher contributions from both agents to maintain balance.

7.2 Consequences of the Changes

Reduction of Free-Riding : *Free-Riding in the Standard Model:* In the original sequential contribution model of public goods, free-riding is a common problem. Agents tend to delay or minimize their contributions, hoping the other agent will contribute more to the public good. *Effect of Altruism:* By incorporating altruism, agent 1 becomes less inclined to free-ride, as their utility is directly linked to agent 2's welfare. This interdependence can lead to higher overall contributions, as both agents are motivated by mutual benefit. *Effect of Warm-Glow:* The warm-glow effect further reduces free-riding. Since agent 1 derives personal satisfaction from contributing (independent of agent 2's contributions), they are less likely to delay their contribution. Instead, they actively participate in the provision of the public good for the sake of their own utility.

Symmetry in Contributions : *Balance in Contributions:* With the introduction of reference dependency, agents are motivated to contribute symmetrically. The disutility term penalizes agent 1 if they contribute less than agent 2, which leads to an equilibrium where both agents tend to contribute similar amounts. This mechanism counteracts free-riding and encourages fairer distribution of the contribution burden. *Effect on Sequential Contribution:* In a sequential setting, the second-moving agent (agent 2) may still adjust their contribution in response to agent 1's decision, but reference dependency encourages agent 1 to contribute more from the start, reducing strategic delays

and enhancing overall efficiency.

Convex vs. Concave Warm-Glow Effects : *Convex Warm-Glow*: When the warm-glow utility is convex (e.g., $\omega(g_1) = g_1^2$), agent 1 experiences increasing returns to contributing more. This may lead to extreme behavior where agent 1 provides the majority, if not all, of the public good. However, this is typically an unrealistic outcome and not widely analyzed in the model. *Concave Warm-Glow*: In the case of concave warm-glow (e.g., $\omega(g_1) = \sqrt{g_1}$), agent 1's marginal utility from contributing decreases as contributions increase, which balances contributions more evenly across both agents. This concave structure helps to create a more realistic scenario where both agents contribute to the public good, but no single agent over-commits.

7.3 Summary of Optimal Contribution

In the modified model with altruism, warm-glow, and reference dependency, the optimal contribution g_1^* for agent 1 is derived by balancing the direct utility from the public good, altruistic utility from agent 2's welfare, and the intrinsic satisfaction from contributing. The analysis shows that agent 1 will tend to contribute more than in the standard model, driven by the combined effects of altruism and warm-glow, as well as the penalty from reference dependency if they contribute less than agent 2.

In conclusion, the modifications introduced to the sequential contribution game—specifically, altruism, warm-glow effects, and reference dependency—lead to higher contributions from both agents, reduce the impact of free-riding, and promote more balanced contributions. This has important implications for real-world public goods scenarios, where encouraging contributions from individuals can lead to more efficient and equitable outcomes.

References

- Admati, Anat R and Motty Perry**, “Joint Projects without Commitment,” *The Review of Economic Studies*, 1991, 58 (2), 259–276.
- Andreoni, James**, “Impure Altruism and Donations to Public Goods: A Theory of Warm-Glow Giving,” *The Economic Journal*, 1990, 100 (401), 464–477.
- Buchholz, Wolfgang and Todd Sandler**, “Global Public Goods: A Survey,” *Journal of Economical Literature*, 2021, 59, 448–545.
- DellaVigna, Stefano, John A List, and Ulrike Malmendier**, “Testing for Altruism and Social Pressure in Charitable Giving,” *Econometrica*, 2012, 80 (2), 747–782.
- Falkinger, Josef, Ernst Fehr, Simon Gächter, and Rudolf Winter-Ebmer**, “Appropriating returns from innovations in public goods projects,” *The American Economic Review*, 2002, 92 (1), 92–113.
- Frey, Bruno S. and Stephan Meier**, “Motivations to Contribute to Public Goods: Beyond Rational Choice Economics,” *Business Strategy and the Environment*, 2004, 13 (4), 209–222.
- Groves, Theodore and John Ledyard**, “Optimal Allocation of Public Goods: A Solution to the ”Free Rider” Problem,” *Econometrica*, 1977, 45 (4), 783–809.
- K., Holt Charles A. Goeree Jacob and Susan K. Laury**, “A Theoretical Analysis of Altruism and Decision Error in Public Goods Games,” *Journal of Public Economics*, 1999, 73 (2), 257–274.
- Kahneman, Daniel and Amos Tversky**, “Prospect Theory: An Analysis of Decision under Risk,” *Econometrica*, 1979, 47 (2), 263–291.

- Kamas, Linda and Anne Preston**, “Distributive and reciprocal fairness: What can we learn from the heterogeneity of social preferences?,” *Journal of Economic Psychology*, 2012, *33*, 538–553.
- Khalil, Elias L.**, “What is altruism?,” *Journal of Economic Psychology*, 2008, *29* (1), 94–116.
- Lilley, Andrew and Robert Slonim**, “The price of warm glow,” *Journal of Public Economics*, 2013, *114* (1), 58–74.
- O’Donoghue, Ted and Charles Sprenger**, “Reference-Dependent Preferences,” *Annual Review of Economics*, 2021, *13*, 431–454.
- Palfrey, Thomas R. and Jeffrey E. Prisbrey**, “Anomalous Behavior in Public Goods Experiments: How Much and Why?,” *The American Economic Review*, 1997, *87* (5), 829–846.
- Varian, Hal R.**, “Sequential contributions to public goods,” *Journal of Public Economics*, 1994, *53* (2), 165–186.

Selbständigkeitserklärung

Ich erkläre hiermit, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen benutzt habe. Alle Stellen, die wörtlich oder sinngemäss aus Quellen entnommen wurden, habe ich als solche gekennzeichnet. Mir ist bekannt, dass andernfalls der Senat gemäss Artikel 36 Absatz 1 Buchstabe o des Gesetzes vom 5. September 1996 über die Universität zum Entzug des aufgrund dieser Arbeit verliehenen Titels berechtigt ist.

Bern, 11.11.2024

(Unterschrift)